

Potassium

Fact Sheet for Health Professionals

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Introduction

Potassium, the most abundant intracellular cation, is an essential nutrient that is naturally present in many foods and available as a dietary supplement. Potassium is present in all body tissues and is required for normal cell function because of its role in maintaining intracellular fluid volume and transmembrane electrochemical gradients [1,2]. Potassium has a strong relationship with sodium, the main regulator of extracellular fluid volume, including plasma volume.

The total amount of potassium in the adult body is about 45 millimole (mmol)/kg body weight (about 140 g for a 175 pound adult; 1 mmol = 1 milliequivalent [mEq] or 39.1 mg potassium) [3]. Most potassium resides intracellularly, and a small amount is in extracellular fluid [2-4]. The intracellular concentration of potassium is about 30 times higher than the extracellular concentration, and this difference forms a transmembrane electrochemical gradient that is maintained via the sodium-potassium (Na⁺/K⁺) ATPase transporter [4]. In addition to maintaining cellular tonicity, this gradient is required for proper nerve transmission, muscle contraction, and kidney function.

Potassium is absorbed via passive diffusion, primarily in the small intestine [2,4,5]. About 90% of ingested potassium is absorbed and used to maintain its normal

intracellular and extracellular concentrations [3,5]. Potassium is excreted primarily in the urine, some is excreted in the stool, and a very small amount is lost in sweat. The kidneys control potassium excretion in response to changes in dietary intakes, and potassium excretion increases rapidly in healthy people after potassium consumption, unless body stores are depleted [2,6]. The kidneys can adapt to variable potassium intakes in healthy individuals, but a minimum of 5 mmol (about 195 mg) potassium is excreted daily in urine [3]. This, combined with other obligatory losses, suggests that potassium balance cannot be achieved with intakes less than about 400–800 mg/day.

Normal serum concentrations of potassium range from about 3.6 to 5.0 mmol/L and are regulated by a variety of mechanisms [3,7]. Diarrhea, vomiting, kidney disease, use of certain medications, and other conditions that alter potassium excretion or cause transcellular potassium shifts can cause hypokalemia (serum levels below 3.6 mmol/L) or hyperkalemia (serum levels above 5.0 mmol/L) [3,5,7,8]. Otherwise, in healthy individuals with normal kidney function, abnormally low or high blood levels of potassium are rare.

Assessing potassium status is not routinely done in clinical practice, and it is difficult to do because most potassium in the body is inside cells. Although blood potassium levels can provide some indication of potassium status, they often correlate poorly with tissue potassium stores [3,9,10]. Other methods to measure potassium status include collecting balance data (measuring net potassium retention and loss); measuring the total amount of potassium or the total amount of exchangeable potassium in the body; and conducting tissue analyses (e.g., muscle biopsies), but all have limitations [9].

Recommended Intakes

Intake recommendations for potassium and other nutrients are provided in the Dietary Reference Intakes (DRIs) developed by expert committees of the National Academies of Sciences, Engineering, and Medicine (NASEM) [11]. DRI is the general term for a set of reference values used for planning and assessing nutrient intakes of healthy people. These values, which vary by age and sex, include the following:

- Recommended Dietary Allowance (RDA): Average daily level of intake sufficient to meet the nutrient requirements of nearly all (97%–98%) healthy individuals; often used to plan nutritionally adequate diets for individuals
- Adequate Intake (AI): Intake at this level is assumed to ensure nutritional adequacy; established when evidence is insufficient to develop an RDA
- Estimated Average Requirement (EAR): Average daily level of intake estimated to meet the requirements of 50% of healthy individuals; usually used to assess the nutrient intakes of groups of people and to plan nutritionally adequate diets for them; can also be used to assess the nutrient intakes of individuals
- Tolerable Upper Intake Level (UL): Maximum daily intake unlikely to cause adverse health effects

In 2019, a NASEM committee updated the DRIs for potassium (and sodium) [11]. The committee found the data insufficient to derive an EAR for potassium. Therefore, they established AIs for all ages based on the highest median potassium intakes in healthy children and adults and on estimates of potassium intakes from breast milk and complementary foods in infants. Table 1 lists the current AIs for potassium for healthy individuals.

Table 1: Adequate Intakes (AIs) for Potassium* [11]

Age	Male	Female	Pregnancy	Lactation
Birth to 6 months	400 mg	400 mg		
7–12 months	860 mg	860 mg		
1–3 years	2,000 mg	2,000 mg		
4–8 years	2,300 mg	2,300 mg		
9–13 years	2,500 mg	2,300 mg		
14–18 years	3,000 mg	2,300 mg	2,600 mg	2,500 mg
19–50 years	3,400 mg	2,600 mg	2,900 mg	2,800 mg
51+ years	3,400 mg	2,600 mg		

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*The AIs do not apply to individuals with impaired potassium excretion because of medical conditions (e.g., kidney disease) or the use of medications that impair potassium excretion.

The NASEM committee also used an expanded DRI model to include a recommended intake level for a nutrient to reduce the risk of chronic disease, what they termed the chronic disease risk reduction intake (CDRR) [11,12]. According to the model, a CDRR might be set for a nutrient like potassium when there is a causal relationship between a certain level of intake and a reduced risk of chronic disease based on evidence of at least moderate strength. However, the committee found the evidence to be insufficient to derive a CDRR for potassium.

Sources of Potassium

Food

Potassium is found in a wide variety of plant and animal foods and in beverages. Many fruits and vegetables are excellent sources, as are some legumes (e.g., soybeans) and potatoes. Meats, poultry, fish, milk, yogurt, and nuts also contain potassium [3,5]. Among starchy foods, whole-wheat flour and brown rice are much higher in potassium than their refined counterparts, white wheat flour and white rice [13].

Milk, coffee, tea, other nonalcoholic beverages, and potatoes are the top sources of potassium in the diets of U.S. adults [14]. Among children in the United States, milk, fruit juice, potatoes, and fruit are the top sources [15].

It is estimated that the body absorbs about 85%–90% of dietary potassium [1,2]. The forms of potassium in fruits and vegetables include potassium phosphate, sulfate, citrate and others but not potassium chloride (the form used in salt substitutes and some dietary supplements; see supplements section below) [16].

Selected food sources of potassium are listed in Table 2.

Table 2: Potassium Content of Selected Foods [13]

Food	Milligrams (mg) per serving	Percent DV*
Apricots, dried, ½ cup	755	16
Lentils, cooked, 1 cup	731	16
Squash, acorn, mashed, 1 cup	644	14

Food	Milligrams (mg) per serving	Percent DV*
Prunes, dried, ½ cup	635	14
Raisins, ½ cup	618	13
Potato, baked, flesh only, 1 medium	610	13
Kidney beans, canned, 1 cup	607	13
Orange juice, 1 cup	496	11
Soybeans, mature seeds, boiled, ½ cup	443	9
Banana, 1 medium	422	9
Milk, 1%, 1 cup	366	8
Spinach, raw, 2 cups	334	7
Chicken breast, boneless, grilled, 3 ounces	332	7
Yogurt, fruit variety, nonfat, 6 ounces	330	7
Salmon, Atlantic, farmed, cooked, 3 ounces	326	7
Beef, top sirloin, grilled, 3 ounces	315	7
Molasses, 1 tablespoon	308	7
Tomato, raw, 1 medium	292	6
Soymilk, 1 cup	287	6
Yogurt, Greek, plain, nonfat, 6 ounces	240	5
Broccoli, cooked, chopped, ½ cup	229	5
Cantaloupe, cubed, ½ cup	214	5
Turkey breast, roasted, 3 ounces	212	5
Asparagus, cooked, ½ cup	202	4
Apple, with skin, 1 medium	195	4
Cashew nuts, 1 ounce	187	4
Rice, brown, medium grain, cooked, 1 cup	154	3
Tuna, light, canned in water, drained, 3 ounces	153	3
Coffee, brewed, 1 cup	116	2
Lettuce, iceberg, shredded, 1 cup	102	2
Peanut butter, 1 tablespoon	90	2
Tea, black, brewed, 1 cup	88	2

Food	Milligrams (mg) per serving	Percent DV*
Flaxseed, whole, 1 tablespoon	84	2
Bread, whole wheat, 1 slice	81	2
Egg, 1 large	69	1
Rice, white, medium grain, cooked, 1 cup	54	1
Bread, white, 1 slice	37	1
Cheese, mozzarella, part skim, 1½ ounces	36	1
Oil (olive, corn, canola, or soybean), 1 tablespoon	0	0

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*DV = Daily Value. The U.S. Food and Drug Administration (FDA) developed DVs to help consumers compare the nutrient contents of foods and dietary supplements within the context of a total diet. The DV for potassium is 4,700 mg for adults and children age 4 years and older [17]. FDA requires the new food labels to list potassium content. Foods providing 20% or more of the DV are considered to be high sources of a nutrient, but foods providing lower percentages of the DV also contribute to a healthful diet.

The U.S. Department of Agriculture's (USDA's) [FoodData Central](https://fdc.nal.usda.gov/) (<https://fdc.nal.usda.gov/>) [13] lists the nutrient content of many foods and provides a comprehensive [list of foods containing potassium](https://fdc.nal.usda.gov/food-search?component=1092) (<https://fdc.nal.usda.gov/food-search?component=1092>). The 2020–2025 [Dietary Guidelines for Americans](https://www.dietaryguidelines.gov/) (<https://www.dietaryguidelines.gov/>) also provides [a list of foods containing potassium](https://www.dietaryguidelines.gov/food-sources-potassium) (<https://www.dietaryguidelines.gov/food-sources-potassium>).

Dietary supplements

In dietary supplements, potassium is often present as potassium chloride, but many other forms—including potassium citrate, phosphate, aspartate, bicarbonate, and gluconate—are also used [18]. The Supplement Facts panel on a dietary supplement label declares the amount of elemental potassium in the product, not the weight of the entire potassium-containing compound. Some dietary supplements contain potassium iodide in microgram amounts, but this ingredient serves as a form of the mineral iodine, not potassium.

Not all multivitamin/mineral supplements contain potassium, but those that do typically provide about 80 mg potassium [18]. Potassium-only supplements are also available, and most contain up to 99 mg potassium. Information on many dietary supplements that contain potassium is available in the [Dietary Supplement Label Database](http://dslid.od.nih.gov) (<http://dslid.od.nih.gov>) from the National Institutes of Health, which contains label information from tens of thousands of dietary supplement products on the market.

Many dietary supplement manufacturers and distributors limit the amount of potassium in their products to 99 mg (which is only about 2% of the DV) because of two concerns related to potassium-containing drugs. First, FDA has ruled that some oral drug products that contain potassium chloride and provide more than 99 mg potassium are not safe because they have been associated with small-bowel lesions [19]. Second, FDA requires some potassium salts containing more than 99 mg potassium per tablet to be labeled with a warning about the reports of small-bowel lesions [20,21]. In accordance with a ruling by Congress, FDA may not limit the amount of any nutrient, including potassium, in a dietary supplement, except for safety-related reasons [22]. However, FDA has not issued a ruling about whether dietary supplements containing more than 99 mg potassium must carry a warning label [21,23].

Only a few studies have examined how well the various forms of potassium in dietary supplements are absorbed. A 2016 dose-response trial found that humans absorb about 94% of potassium gluconate in supplements, and this absorption rate is similar to that of potassium from potatoes [24]. According to an older study, liquid forms of potassium chloride (used as drugs to treat conditions such as digitalis intoxication or arrhythmias due to hypokalemia) are absorbed within a few hours [6]. Enteric coated tablet forms of potassium chloride (designed to prevent dissolution in the stomach but allow it in the small intestine) are not absorbed as rapidly as liquid forms [25].

Salt substitutes

Many salt substitutes contain potassium chloride as a replacement for some or all of the sodium chloride in salt. The potassium content of these products varies widely, from about 440 mg to 2,800 mg potassium per teaspoon [1]. Some people,

such as those with kidney disease or who are taking certain medications, should consult their health care provider before taking salt substitutes because of the risk of hyperkalemia posed by the high levels of potassium in these products.

Potassium Intakes and Status

Dietary surveys consistently show that people in the United States consume less potassium than recommended, which is why the 2015–2020 *Dietary Guidelines for Americans* identifies potassium as a nutrient of public health concern [26].

According to data from the 2013–2014 National Health and Nutrition Examination Survey (NHANES), the average daily potassium intake from foods is 2,423 mg for males age 2–19, and 1,888 mg for females age 2–19 [27]. In adults age 20 and over, the average daily potassium intake from foods is 3,016 mg for men and 2,320 mg for women.

Average potassium intakes vary by race. Non-Hispanic blacks age 20 and older consume an average of 2,449 mg potassium per day. Average daily intakes are 2,695 mg for Hispanic whites and 2,697 mg for non-Hispanic whites [27].

Use of potassium-containing dietary supplements does not significantly increase total potassium intakes among U.S. adults [28], probably because most potassium-containing dietary supplements provide no more than 99 mg potassium per serving [21]. Data from NHANES 2013–2014 indicate that 12% of children and adults age 2 and over use supplements containing potassium, and among those who do, supplement use adds a mean of only 87 mg to total daily potassium intakes [27].

Potassium Deficiency

Insufficient potassium intakes can increase blood pressure, kidney stone risk, bone turnover, urinary calcium excretion, and salt sensitivity (meaning that changes in sodium intakes affect blood pressure to a greater than normal extent) [1].

Severe potassium deficiency can cause hypokalemia, (serum potassium level less than about 3.6 mmol/L) [3,7,8]. Hypokalemia affects up to 21% of hospitalized patients, usually because of the use of diuretics and other medications [29,30], but it is rare among healthy people with normal kidney function.

Mild hypokalemia is characterized by constipation, fatigue, muscle weakness, and malaise [3]. Moderate to severe hypokalemia (serum potassium level less than about 2.5 mmol/L) can cause polyuria (large volume of dilute urine); encephalopathy in patients with kidney disease; glucose intolerance; muscular paralysis; poor respiration; and cardiac arrhythmias, especially in individuals with underlying heart disease [1,3,7]. Severe hypokalemia can be life threatening because of its effects on muscle contraction and, hence, cardiac function [5].

Hypokalemia is rarely caused by low dietary potassium intake alone, but it can result from diarrhea due to potassium losses in the stool. It can also result from vomiting, which produces metabolic alkalosis, leading to potassium losses in the kidneys. Hypokalemia can also be caused by refeeding syndrome (the metabolic response to initial refeeding after a starvation period) because of potassium's movement into cells, laxative abuse, diuretic use, eating clay (a type of pica), heavy sweating, or dialysis [3,5,7,31,32].

Magnesium depletion can contribute to hypokalemia by increasing urinary potassium losses [1,33,34]. It can also increase the risk of cardiac arrhythmias by decreasing intracellular potassium concentrations. More than 50% of individuals with clinically significant hypokalemia might have magnesium deficiency [34]. In people with hypomagnesemia and hypokalemia, both should be treated concurrently [7].

Groups at Risk of Potassium Inadequacy

Potassium inadequacy can occur with intakes that are below the AI but above the amount required to prevent hypokalemia. The following groups are more likely than others to have poor potassium status.

People with inflammatory bowel diseases

Potassium is secreted within the colon, and this process is normally balanced by absorption [35]. However, in inflammatory bowel disease (including Crohn's disease and ulcerative colitis), potassium secretion increases, which can lead to poor potassium status. Inflammatory bowel diseases are also characterized by chronic diarrhea, which can further increase potassium excretion [36].

People who use certain medications, including diuretics and laxatives

Certain diuretics (e.g., thiazide diuretics) that are commonly used to treat high blood pressure increase urinary potassium excretion and can cause hypokalemia [7,8]. Potassium-sparing diuretics, however, do not increase potassium excretion and can actually cause hyperkalemia. Large doses of laxatives and repeated use of enemas can also cause hypokalemia because they increase losses of potassium in stool.

People with pica

Pica is the persistent eating of non-nutritive substances, such as clay. When consumed, clay binds potassium in the gastrointestinal tract, which can increase potassium excretion and lead to hypokalemia [5,31,32]. Cessation of pica combined with potassium supplementation can restore potassium status and resolve symptoms of potassium deficiency.

Potassium and Health

Because of potassium's wide-ranging roles in the body, low intakes can increase the risk of illness. This section focuses on four diseases and disorders in which potassium might be involved: hypertension and stroke, kidney stones, bone health, and blood glucose control and type 2 diabetes.

Hypertension and stroke

Hypertension, a major risk factor for coronary heart disease and stroke, affects almost a third of Americans [2,37]. According to an extensive body of literature, low potassium intakes increase the risk of hypertension, especially when combined with high sodium intakes [16,38-40]. Higher potassium intakes, in contrast, may help decrease blood pressure, in part by increasing vasodilation and urinary sodium excretion, which in turn reduces plasma volume [1]; this effect may be most pronounced in salt-sensitive individuals [2,3,5,37,41].

The Dietary Approaches to Stop Hypertension (DASH) eating pattern, which emphasizes potassium from fruits, vegetables, and low-fat dairy products, lowers

systolic blood pressure by an average of 5.5 mmHg and diastolic blood pressure by 3.0 mmHg [42]. The DASH eating pattern provides three times more potassium than the average American diet. However, it also increases intakes of other nutrients, such as magnesium and calcium, that are also associated with reductions in blood pressure, so potassium's independent contribution cannot be determined. Additional information and sample DASH menu plans are available on the National Heart, Lung, and Blood Institute website. (<https://www.nhlbi.nih.gov/health-topics/dash-eating-plan>).

Results from most clinical trials suggest that potassium supplementation reduces blood pressure. A 2017 meta-analysis of 25 randomized controlled trials in 1,163 participants with hypertension found significant reductions in systolic blood pressure (by 4.48 mm Hg) and diastolic blood pressure (by 2.96 mmHg) with potassium supplementation, mostly as potassium chloride at 30–120 mmol/day potassium (1,173–4,692 mg), for 4–15 weeks [43]. Another meta-analysis of 15 randomized controlled trials found that potassium supplements (mostly containing potassium chloride at 60–65 mEq/day potassium [2,346–2,541 mg]) for 4–24 weeks in 917 patients with normal blood pressure or hypertension who were not taking antihypertensive medications significantly reduced both systolic and diastolic blood pressure [44]. The supplements had the greatest effect in patients with hypertension, reducing systolic blood pressure by a mean of 6.8 mmHg and diastolic blood pressure by 4.6 mmHg. Two earlier meta-analyses of 19 trials [45] and 33 trials [46] had similar findings. However, a Cochrane Review of six of the highest-quality trials found nonsignificant reductions in systolic and diastolic blood pressure with potassium supplementation [47].

In 2018, the Agency for Healthcare Research and Quality (AHRQ) published a systematic review of the effects of sodium and potassium intakes on chronic disease outcomes and their risk factors [48]. The authors concluded that, based on observational studies, the associations between dietary potassium intakes and lower blood pressure in adults were inconsistent. They also found no evidence for an association between potassium intakes and the risk of hypertension. The authors did report, however, that potassium supplements (mostly containing potassium chloride) in doses ranging from 20 to 120 mmol/day (782 to 4,692 mg/day) for 1 to 36 months lowered both systolic and diastolic blood pressure compared to placebo. A similar analysis conducted by the NASEM committee that

included 16 trials found that potassium supplements significantly lowered systolic blood pressure by a mean of 6.87 mmHg and diastolic blood pressure by 3.57 mmHg [11]. However, the effects were stronger among studies including participants with hypertension; for studies including only participants without hypertension, the effects were not statistically significant. Based on 13 randomized controlled trials that primarily enrolled patients with hypertension, the AHRQ review found that the use of potassium-containing salt substitutes in place of sodium chloride significantly reduced systolic blood pressure in adults by a mean of 5.58 mmHg and diastolic blood pressure by 2.88 mmHg [48]. However, reducing sodium intake decreased both systolic and diastolic blood pressure in adults, and increasing potassium intake via food or supplements did not reduce blood pressure any further. This finding suggests that at least some of the beneficial effects of potassium salt substitutes on blood pressure may be due to the accompanying reduction in sodium intake, rather than the increase in potassium intake.

Higher potassium intakes have been associated with a decreased risk of stroke and possibly other cardiovascular diseases (CVDs) [16,49]. A meta-analysis of 11 prospective cohort studies in 247,510 adults found that a 1,640 mg per day higher potassium intake was associated with a significant 21% lower risk of stroke as well as nonsignificant lower risks of coronary heart disease and total CVD [39]. Similarly, the authors of a meta-analysis of nine cohort studies reported a significant 24% lower risk of stroke with higher potassium intakes and a nonsignificant reduction in coronary heart disease and CVD risk [50]. However, the AHRQ review found inconsistent relationships between potassium intakes and risk of stroke based on 15 observational studies [48].

Any beneficial effect of potassium on CVD is likely due to its antihypertensive effects. However, some research shows a benefit even when blood pressure is accounted for. For example, a 2016 meta-analysis of 16 cohort studies with a total of 639,440 participants found that those with the highest potassium intakes (median 103 mmol [4,027 mg] per day) had a 15% lower risk of stroke than those with the lowest potassium intakes (median 52.5 mmol [2,053 mg] per day). In addition, participants who consumed 90 mmol potassium/day (approximately 3,500 mg) had the lowest risk of stroke [51]. However, even when blood pressure was accounted for, higher potassium intakes still produced a significant 13% lower

risk of stroke. These findings suggest that other mechanisms (e.g., improved endothelial function and reduced free radical formation) may be involved [37].

FDA has approved the following health claim: “Diets containing foods that are a good source of potassium and that are low in sodium may reduce the risk of high blood pressure and stroke” [17]. Overall, the evidence suggests that consuming more potassium might have a favorable effect on blood pressure and stroke, and it might also help prevent other forms of CVD. However, more research on both dietary and supplemental potassium is needed before firm conclusions can be drawn.

Kidney stones

Kidney stones are most common in people age 40 to 60 [52]. Stones containing calcium—in the form of calcium oxalate or calcium phosphate—are the most common type of kidney stone. Low potassium intakes impair calcium reabsorption within the kidney, increasing urinary calcium excretion and potentially causing hypercalciuria and kidney stones [16,37]. Low urinary levels of citrate also contribute to kidney stone development.

Observational studies show inverse associations between dietary potassium intakes and risk of kidney stones. In a cohort of 45,619 men age 40 to 75 years with no history of kidney stones, those with the highest potassium intakes ($\geq 4,042$ mg/day on average) had a 51% lower risk of kidney stones over 4 years of follow-up than those with the lowest intakes ($\leq 2,895$ mg/day) [53]. Similarly, in more than 90,000 women age 34–59 who participated in the Nurses’ Health Study and had no history of kidney stones, those who consumed an average of more than 4,099 mg of potassium per day had a 35% lower risk of kidney stones over a 12-year follow-up period than those who averaged less than 2,407 mg of potassium per day [54].

Some research suggests that supplementation with potassium citrate reduces hypercalciuria as well as the risk of kidney stone formation and growth [52,55]. In a clinical trial of 57 patients with at least two kidney stones (either calcium oxalate or calcium oxalate plus calcium phosphate) over the previous 2 years and hypocitraturia (low urinary citrate levels), supplementation with 30–60 mEq potassium citrate (providing 1,173 to 2,346 mg potassium) for 3 years significantly

reduced kidney stone formation compared with placebo [55]. This study was included in a 2015 Cochrane Review of seven studies that examined the effects of potassium citrate, potassium-sodium citrate, and potassium-magnesium citrate supplementation on the prevention and treatment of calcium-containing kidney stones in a total of 477 participants, most of whom had calcium oxalate stones [52]. The potassium citrate salts significantly reduced the risk of new stones and reduced stone size. However, the proposed mechanism involves citrate, not potassium per se; citrate forms complexes with urinary calcium and increases urine pH, inhibiting the formation of calcium oxalate crystals [52,56]. The authors of the AHRQ review [48] concluded that observational studies suggest an association between higher potassium intakes and lower risk of kidney stones. However, they also found the evidence insufficient to determine whether potassium supplements are effective because only one trial that addressed this question [55] met their inclusion criteria.

Additional research is needed to fully understand the potential link between dietary and supplemental potassium and the risk of kidney stones.

Bone health

Observational studies suggest that increased consumption of potassium from fruits and vegetables is associated with increased bone mineral density [57]. This evidence, combined with evidence from metabolic studies and a few clinical trials, suggests that dietary potassium may improve bone health.

The underlying mechanisms are unclear, but one hypothesis is that potassium helps protect bone through its effect on acid-base balance [37]. Diets that are high in acid-forming foods, such as meats and cereal grains, contribute to metabolic acidosis and might have an adverse effect on bone. Alkaline components in the form of potassium salts (potassium bicarbonate or citrate, but not potassium chloride) from food or potassium supplements might counter this effect and help preserve bone tissue. In the Framingham Heart Study for example, higher potassium intake was associated with significantly greater bone mineral density in 628 elderly men and women [58]. In another study, the DASH eating pattern significantly reduced biochemical markers of bone turnover [59]. This eating pattern has a lower acid load than typical Western diets and is also high in calcium

and magnesium, in addition to potassium, so any independent contribution of potassium cannot be determined.

Only a few clinical trials have examined the effects of potassium supplements on markers of bone health. One trial found that supplementation with potassium citrate at either 60 mmol/day (2,346 mg potassium) or 90 mmol/day (3,519 mg potassium) for 6 months significantly reduced urinary calcium excretion compared with placebo in 52 healthy men and women older than 55 years [60]. In another clinical trial, 201 healthy adults age 65 years or older received daily supplementation with 60 mEq potassium citrate (providing 2,346 mg potassium) or placebo as well as 500 mg/day calcium (as calcium carbonate) and 400 International Units (IU)/day vitamin D3 for 2 years [61]. Potassium supplementation significantly increased bone mineral density at the lumbar spine and bone microarchitecture compared with placebo. In a similar clinical trial among older adults, supplemental potassium bicarbonate (mean doses of 2,893 or 4,340 mg/day potassium) for 84 days significantly reduced biochemical markers of bone turnover and urinary calcium excretion [62]. Conversely, a clinical trial in 276 postmenopausal women age 55–65 years found that supplementation with potassium citrate at either 18.5 mEq/day (providing 723 mg potassium) or 55.5 mEq/day (2,170 mg potassium) for 2 years did not significantly reduce bone turnover or increase bone mineral density at the hip or lumbar spine compared with placebo [63].

Overall, higher intakes of potassium from diets that emphasize fruits and vegetables might improve bone health. However, more research is needed to elucidate the underlying mechanisms and tease out potassium's individual contribution.

Blood glucose control and type 2 diabetes

Type 2 diabetes is a growing public health concern that currently affects almost 12% of U.S. adults [64]. Although individuals with obesity have an elevated risk of type 2 diabetes, other metabolic factors also play a role. Because potassium is needed for insulin secretion from pancreatic cells, hypokalemia impairs insulin secretion and could lead to glucose intolerance [2]. This effect has been observed mainly with long-term use of diuretics (particularly those containing thiazides) or

hyperaldosteronism (excessive aldosterone production), which both increase urinary potassium losses, but it can occur in healthy individuals as well [2,10,16,65].

Numerous observational studies of adults have found associations between lower potassium intakes or lower serum or urinary potassium levels and increased rates of fasting glucose, insulin resistance, and type 2 diabetes [66-72]. These associations might be stronger in African Americans, who tend to have lower potassium intakes, than in whites [68,71]. For example, one study of 1,066 adults age 18–30 years without diabetes found that those with urinary potassium levels in the lowest quintile were more than twice as likely to develop type 2 diabetes over 15 years of follow-up than those in the highest quintile [68]. Among 4,754 participants from the same study with potassium intake data, African Americans with lower potassium intakes had a significantly greater risk of type 2 diabetes over 20 years of follow-up than those with higher intakes, but this association was not found in whites.

In another observational study, which analyzed data from 84,360 women age 34–59 years participating in the Nurses' Health Study, those in the highest quintile of potassium intake had a 38% lower risk of developing type 2 diabetes over 6 years of follow-up than those in the lowest quintile [66]. Serum potassium levels were inversely associated with fasting glucose levels in 5,415 participants age 45–84 years from the Multi-Ethnic Study of Atherosclerosis, but these levels had no significant association with diabetes risk over 8 years of follow-up [70].

Although observational studies suggest that potassium status is linked to blood glucose control and type 2 diabetes, this association has not been adequately evaluated in clinical trials. In a small clinical trial in 29 African American adults with prediabetes and low to normal serum potassium levels (3.3–4.0 mmol/L), supplementation with 40 mEq (1,564 mg) potassium (as potassium chloride) for 3 months significantly lowered fasting glucose levels, but it did not affect glucose or insulin measures during an oral glucose tolerance test [73].

The findings from studies conducted to date are promising. However, more research, including randomized controlled trials, is needed before potassium's link with blood glucose control and type 2 diabetes can be confirmed.

Health Risks from Excessive Potassium

Dietary potassium

In healthy people with normal kidney function, high dietary potassium intakes do not pose a health risk because the kidneys eliminate excess amounts in the urine [1]. Although case reports indicate that very large doses of potassium supplements can cause heart abnormalities and death, the NASEM committee concluded that these reports do not provide sufficient evidence to set a UL [11]. In addition, there is no evidence that high intakes of potassium cause hyperkalemia in adults with normal kidney function or other adverse effects. Therefore, the committee did not set a UL for potassium.

However, in people with impaired urinary potassium excretion due to chronic kidney disease or the use of certain medications, such as angiotensin converting enzyme (ACE) inhibitors or potassium-sparing diuretics, even dietary potassium intakes below the AI can cause hyperkalemia [11]. Hyperkalemia can also occur in people with type 1 diabetes, congestive heart failure, adrenal insufficiency, or liver disease [7]. Individuals at risk of hyperkalemia should consult a physician or registered dietitian about appropriate potassium intakes from all sources. Information on low-potassium diets is also available from the National Kidney Disease Education Program. (<https://www.niddk.nih.gov/health-information/professionals/-/media/Files/Health-Information/Communication-Programs/NKDEP/nutrition-potassium-508.pdf>).

Although hyperkalemia can be asymptomatic, severe cases can cause muscle weakness, paralysis, heart palpitations, paresthesias (a burning or prickling sensation in the extremities), and cardiac arrhythmias that could be life threatening [1,7].

Potassium from dietary supplements, salt substitutes, and medications

Potassium supplements can cause minor gastrointestinal side effects [48]. Chronic ingestion of doses of potassium supplements (e.g., up to 15,600 mg for 5 days) in healthy people can increase plasma levels of potassium, but not beyond the normal

range [1]. However, very high amounts of potassium supplements or salt substitutes that contain potassium could exceed the kidney's capacity to excrete potassium, causing acute hyperkalemia even in healthy individuals.

The use of potassium salts in certain medications has been associated with small-bowel lesions, causing obstruction, hemorrhage, and perforation [20,74]. For this reason, FDA requires some oral drugs providing more than 99 mg of potassium to be labeled with a warning.

Interactions with Medications

Several types of medications have the potential to affect potassium status in ways that could be dangerous. A few examples are provided below. People taking these and other medications should discuss their potassium intakes and status with their health care providers.

Angiotensin converting enzyme inhibitors and angiotensin receptor blockers

ACE inhibitors, such as benazepril (Lotensin), and angiotensin receptor blockers (ARBs) such as losartan (Cozaar), are used to treat hypertension and heart failure, slow progression of kidney disease in patients with chronic kidney disease and type 2 diabetes, and decrease morbidity and mortality after myocardial infarction [75-77]. These medications reduce urinary potassium excretion, which can lead to hyperkalemia. Experts recommend monitoring potassium status in people taking ACE inhibitors or ARBs, especially if they have other risk factors for hyperkalemia, such as impaired kidney function [75].

Potassium-sparing diuretics

Potassium-sparing diuretics, such as amiloride (Midamor) and spironolactone (Aldactone), reduce the excretion of potassium in the urine and can cause hyperkalemia [77,78]. Experts recommend monitoring potassium status in people taking these medications, especially if they have impaired kidney function or other risk factors for hyperkalemia [78].

Loop and thiazide diuretics

Treatment with loop diuretics, such as furosemide (Lasix) and bumetanide (Bumex), and thiazide diuretics, such as chlorothiazide (Diuril) and metolazone (Zaroxolyn), increases urinary potassium excretion and can lead to hypokalemia [77,78]. Experts recommend monitoring potassium status in people taking these medications and initiating potassium supplementation if warranted [77].

Potassium and Healthful Diets

In general, a person's nutritional needs should be met primarily through the diet, including fortified foods. Dietary supplements may be useful in cases where it is not possible to meet the needs for specific nutrients through food alone, especially during certain life stages. The *Dietary Guidelines for Americans* offers a general description of healthy dietary patterns.

For more information about building a healthy dietary pattern, refer to the *Dietary Guidelines for Americans* (<https://www.dietaryguidelines.gov>).

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